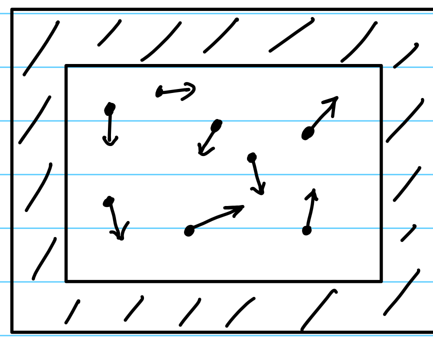
Math Thermo
class # 09
02/17/2026

Statistical Mechanics of Non-Interacting Particle Systems

Consider a system with N particles in a box of volume $V > 0$, isolated from the rest of the universe.



Suppose that each particle can be in any one of exactly $r \in \mathbb{N}$ energy levels.

The tuple

$$(n_1, n_2, \cdots, n_r) \in \mathbb{N}_0^r$$

where

(9.1) $$N = \sum_{i=1}^{r} n_i$$

completely characterizes the system. Here,

$$n_i = \text{number of particles in energy level } i$$

The internal energy of the particles in

configuration $(n_1, n_2, \cdots, n_r)$ is

(9.2) $$U = \sum_{i=1}^{r} n_i U_i ,$$

where U_i is the fixed energy of level i.

The vector

$$\vec{\nu} := (n_1, n_2, \cdots, n_r) \in \mathbb{N}_0^r$$

is called a <u>macrostate</u>.

From combinatorics, we know that the number of ways that N distinguishable particles can be arranged in the macrostate

$$\vec{\nu} := (n_1, n_2, \cdots, n_r) \in \mathbb{N}_0^r$$

is precisely

(9.3) $$W(\vec{\nu}) = \frac{N!}{n_1!\, n_2! \cdots n_r!} .$$

<u>Example</u> (9.1): Suppose that $r = 3$ and $N = 4$, that is, four particles can be put into three energy bins.

	Macrostate, $\vec{\nu}$	$W(\vec{\nu})$
1	$(4,0,0)$	1
2	$(0,4,0)$	1
3	$(0,0,4)$	1

	Macrostate, $\vec{\nu}$	$W(\vec{\nu})$
4	$(3,1,0)$	$\frac{4!}{3!\,1!\,0!} = 4$
5	$(3,0,1)$	4
6	$(1,3,0)$	4
7	$(0,3,1)$	4
8	$(1,0,3)$	4
9	$(0,1,3)$	4
10	$(2,2,0)$	$\frac{4!}{2!\,2!\,0!} = 6$
11	$(2,0,2)$	6
12	$(0,2,2)$	6
13	$(2,1,1)$	$\frac{4!}{2!\,1!\,1!} = 12$
14	$(1,2,1)$	12
15	$(1,1,2)$	12

The total number of distinct macrostates is 15.

The total number of microstates is

$$3(1) + 6(4) + 3(6) + 3(12) = 81 = 3^4.$$

Generically the total number of microstates is r^N.

Example (9.2): Let us examine the 12 distinct microstates of the macrostate $\vec{\nu} = (1,2,1)$ mentioned above. The particles are all of the same type, that is, they are all argon atoms, for example. But we imagine that each has a label to distinguish them, one from another:

1 2 3 4

	1	2	1
microstate	U_1 bin	U_2 bin	U_3 bin
1	1	2 3	4
2	1	2 4	3
3	1	3 4	2
4	2	3 4	1
5	2	1 3	4
6	2	1 4	3
7	3	1 4	2
8	3	2 4	1
9	3	1 2	4
10	4	1 2	3
11	4	1 3	2
12	4	2 3	1

Each microstate in the macrostate $\vec{\nu} = (1, 2, 1)$ has exactly the same energy

$$U(\vec{\nu}) = \sum_{i=1}^{3} n_i U_i = 1U_1 + 2U_2 + 1U_3.$$

Definition (9.3): The Boltzmann (or counting) entropy of the macrostate $\vec{\nu} \in \mathbb{N}_0^r$ is, precisely,

$$S_B(\vec{\nu}) = k_B \log(W(\vec{\nu})),$$

where k_B is Boltzmann's constant

$$k_B \cong 1.38 \times 10^{-23} \text{ J/K}.$$

<u>Definition</u> (9.4): A system of N isolated distinguishable particles of definite energy U, with r levels (energy bins) is called <u>Ergodic</u> iff

1) it is deterministic,

2) the time spent in each microstate, the transition time, $\tau > 0$, is always the same,

3) it is cyclic,

4) and each microstate is visited once and only once per cycle.

<u>Example</u> (9.5) What does this mean? Suppose for a certain number of particles, there are exactly 10 microstates with exactly energy U.

$$\begin{tikzcd}
M_1 \arrow[r, "\tau"] & M_2 \arrow[r, "\tau"] & M_3 \arrow[r, "\tau"] & M_4 \arrow[d, "\tau"] \\
M_{10} \arrow[u, "\tau"] & & & M_5 \arrow[d, "\tau"] \\
M_9 \arrow[u, "\tau"] & M_8 \arrow[l, "\tau"'] & M_7 \arrow[l, "\tau"'] & M_6 \arrow[l, "\tau"']
\end{tikzcd}$$

One cycle is represented above. We imagine that the ~~state~~ sits in microstate M_i for exactly $\tau > 0$ seconds before transitioning to M_{i+1}.

The deterministic assumption means that the sequence of microstates is always the same.

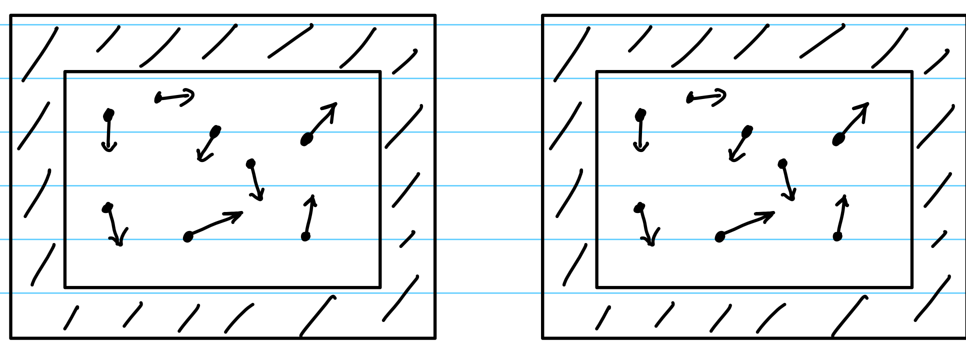
An ergodic system spends the most time in the macrostate with the largest number of microstates.

Definition (9.6): Of all of the macrostates with the same total energy U, the one with the largest number of microstates is called the equilibrium macrostate or just the equilibrium.

Remark: Since the entropy is an increasing function of $W(\vec{\nu})$, the Boltzmann entropy is a maximum for the equilibrium macrostate.

Why use log?

Suppose we have now two identicle boxes of particles.



α $\quad$ β

What is the entropy in this system?

If system α has W_α states and system β has W_β states, the combined system has $W_\alpha \times W_\beta$ states (multiplicative).

But observe

$$\log(W_\alpha \times W_\beta) = \log(W_\alpha) + \log(W_\beta)$$

which is additive!

The Boltzmann entropy is, therefore, additive, that is, extensive.

<u>Theorem</u> (9.7): (Sterling's Approx.)

$$\ln(n!) \sim n\log(n) - n \quad , \text{ as } n \to \infty$$

which means that the relative error becomes small:

$$\frac{\log(n!) - (n\log(n) - n)}{n\log(n) - n} \xrightarrow{n\to\infty} 0 .$$

Using this approximation, we find, as $n_i \to \infty$,

$$\begin{aligned}
\log(W(\vec{\nu})) &= \log(N!) - \sum_{i=1}^{r} \log(n_i!) \\
&\sim N\log(N) - N - \sum_{i=1}^{r} (n_i \log(n_i) - n_i) \\
&= N\log(N) - N - \sum_{i=1}^{r} n_i \log(n_i) + N \\
&= \sum_{i=1}^{r} n_i (\log(N) - \log(n_i)) \\
&= -\sum_{i=1}^{r} n_i \log\left(\frac{n_i}{N}\right).
\end{aligned}$$

Thus,

(9.4) $$S_B(\vec{\nu}) \sim -k_B \sum_{i=1}^{r} n_i \log\left(\frac{n_i}{N}\right), \quad \text{as } n_i \to \infty.$$

Define
$$p_i := \frac{n_i}{N}, \quad i = 1, \dots, r.$$

Then,
$$\sum_{i=1}^{r} p_i = 1.$$

Clearly, p_i represents a probability. But what does the probability mean?

<u>Example</u> (9.8): For $N=4$, $r=3$, consider the macrostate
$$\vec{\nu} = (n_1, n_2, n_3)$$
$$= (2, 1, 1)$$

Then
$$p_1 = \frac{1}{2}, \quad p_2 = \frac{1}{4}, \quad p_3 = \frac{1}{4}.$$

Given an arbitrary particle, the probability that it is in state U_1 is $p_1 = \frac{1}{2}$. The probability it is in state U_3 (energy bin U_3) is $p_3 = \frac{1}{4}$.

<u>Equilibrium Postulate</u>: For an isolated physical system of N particles with fixed total internal energy U, equilibrium is that macrostate $\vec{\nu} = (n_1, \dots, n_r)$, among all macrostates with internal energy U maximizes

the <u>Gibbs entropy</u>

(9.5) $$S_G(\vec{\nu}) = -k_B \sum_{i=1}^{r} n_i \log\left(\frac{n_i}{N}\right).$$

or, equivalently, that maximizes the <u>average Gibbs entropy</u>

(9.6) $$\bar{S}_G(\vec{\nu}) = -k_B \sum_{i=1}^{r} \frac{n_i}{N} \log\left(\frac{n_i}{N}\right) = -k_B \sum_{i=1}^{r} p_i \log(p_i).$$

<u>Theorem</u> (9.9): Suppose that $r, N \in \mathbb{N}$ are fixed. Assume that the constants

(9.7) $$N = \sum_{i=1}^{r} n_i, \qquad n_i \geq 0$$

(9.8) $$U = \sum_{i=1}^{r} U_i n_i$$

hold. The macrostate $\vec{\nu} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$ that maximizes the Gibbs entropy (9.5) is given by

(9.9) $$n_j^{eq} = \exp\left(\frac{-\beta U_j}{k_B}\right) \cdot \frac{N}{Z}$$

where

(9.10) $$Z = \sum_{i=1}^{r} \exp\left(\frac{-\beta U_i}{k_B}\right)$$

is the so-called <u>partition function</u> and β is a positive, unique number that will be determined.

Proof: Let us use the Lagrange Multiplier approach. Set

$$J(\vec{\nu},\alpha,\beta) := \sum_{i=1}^{r} n_i k_B \log\left(\frac{n_i}{N}\right) + \alpha\left(\sum_{i=1}^{r} n_i - N\right) + \beta\left(\sum_{i=1}^{r} n_i U_i - U\right).$$

We wish to find the minimum of J.

Taking derivatives, the minimizer must satisfy

(9.11) $$\frac{\partial J}{\partial n_i} = k_B \log\left(\frac{n_i}{N}\right) + k_B \cancel{n_i} \frac{\cancel{N}}{\cancel{n_i}} \frac{1}{\cancel{N}} + \alpha + \beta U_i = 0,$$

$$\frac{\partial J}{\partial \alpha} = \sum_{i=1}^{r} n_i - N = 0,$$

$$\frac{\partial J}{\partial \beta} = \sum_{i=1}^{r} n_i U_i - U = 0.$$

The last two equations are just our constraints.
Equation (9.11) is equivalent to

$$n_j = N \exp\left(\frac{-(\alpha+k_B)}{k_B}\right) \exp\left(\frac{-\beta U_j}{k_B}\right).$$

Using (9.7)

$$\cancel{N} = \cancel{N} \exp\left(\frac{-(\alpha+k_B)}{k_B}\right) \sum_{i=1}^{r} \exp\left(\frac{-\beta U_i}{k_B}\right)$$

Thus

$$\exp\left(\frac{\alpha+k_B}{k_B}\right) = \sum_{i=1}^{r} \exp\left(\frac{-\beta U_i}{k_B}\right) = Z.$$

If you really want to find α, you can solve for it:

$$\alpha = k_B \log(Z) - k_B.$$

In any case, we have the expression we wanted

$$n_i^{eq} = \frac{N}{Z} \exp\left(\frac{-\beta U_i}{k_B}\right).$$

To determine β, we need to use the energy constraint (9.8):

(9.11) $$\frac{N}{Z} \sum_{i=1}^{r} U_i \exp\left(\frac{-\beta U_i}{k_B}\right) = U.$$

It is not possible to get a generic, closed-form expression for β, but there is a unique solution $\beta > 0$, given U, N, and the energy levels, U_i.

We leave it as an exercise to the reader to prove this. ///

In lieu of a proof of the last claim, we will use two simple examples to show how β may be computed.

What we find is that β will depend upon the internal energy, U, and the number of particles.

Example (9.10): Consider the following two-state system

$$U_1 = 0, \qquad U_2 = \varepsilon > 0.$$

The constraints are, for fixed N and U,

$$n_1 + n_2 = N$$

$$n_1 \cdot 0 + n_2 \cdot \varepsilon = U$$

Thus, n_1 and n_2 are completely determined:

$$n_1^{eq} = N - \frac{U}{\varepsilon}, \qquad n_2^{eq} = \frac{U}{\varepsilon}.$$

We can calculate the Gibbs entropy directly

$$S_G(\vec{\nu}_{eq}) = -k_B \left(N - \frac{U}{\varepsilon}\right) \log\left(1 - \frac{U}{N\varepsilon}\right) - k_B \frac{U}{\varepsilon} \log\left(\frac{U}{N\varepsilon}\right).$$

This is the entropy at equilibrium; there are no free parameters and no maximum to find for fixed U and N.

For the entropy at equilibrium we write

$$S_{eq}(U, N) := S_G(\vec{\nu}_{eq}).$$

Let us calculate the temperature using the usual procedure.

$$\frac{1}{T} = \frac{\partial S_{eq}}{\partial U} = \frac{k_B}{\varepsilon}\log\left(1-\frac{U}{N\varepsilon}\right)$$
$$-k_B\left(N-\frac{U}{\varepsilon}\right)\frac{1}{1-\frac{U}{N\varepsilon}}\left(-\frac{1}{N\varepsilon}\right)$$
$$-\frac{k_B}{\varepsilon}\log\left(\frac{U}{N\varepsilon}\right)$$
$$-\frac{k_B U}{\varepsilon}\,\frac{N\varepsilon}{U}\,\frac{1}{N\varepsilon}$$

$$= \frac{k_B}{\varepsilon}\log\left(\frac{1-\frac{U}{N\varepsilon}}{\frac{U}{N\varepsilon}}\right) + \frac{k_B}{\varepsilon} - \frac{k_B}{\varepsilon}.$$

Thus,

$$\frac{1}{T} = \frac{k_B}{\varepsilon}\log\left(\frac{N\varepsilon}{U}-1\right).$$

Next, let us use the general solution procedure that we developed in Theorem (9.9).

$$n_1^{eq} = \frac{N\exp\left(\frac{-\beta\cdot 0}{k_B}\right)}{\exp\left(-\frac{\beta\cdot 0}{k_B}\right)+\exp\left(\frac{-\beta\cdot\varepsilon}{k_B}\right)} = \frac{N}{1+\exp\left(-\frac{\beta\varepsilon}{k_B}\right)}$$

$$n_2^{eq} = \frac{N\exp\left(\frac{-\beta\cdot\varepsilon}{k_B}\right)}{\exp\left(-\frac{\beta\cdot 0}{k_B}\right)+\exp\left(\frac{-\beta\cdot\varepsilon}{k_B}\right)} = \frac{N\exp\left(-\frac{\beta\varepsilon}{k_B}\right)}{1+\exp\left(-\frac{\beta\varepsilon}{k_B}\right)}$$

The energy constraint is, recall,

$$n_1^{eq} \cdot 0 + n_2^{eq} \varepsilon = U.$$

Thus,

$$n_2^{eq} = \frac{U}{\varepsilon} = \frac{N}{1 + \exp\left(\frac{\beta\varepsilon}{k_B}\right)}$$

Therefore,

$$1 + \exp\left(\frac{\beta\varepsilon}{k_B}\right) = \frac{\varepsilon N}{U}$$

$$\exp\left(\frac{\beta\varepsilon}{k_B}\right) = \frac{\varepsilon N}{U} - 1$$

$$\boxed{\beta = \frac{k_B}{\varepsilon} \log\left(\frac{\varepsilon N}{U} - 1\right).}$$

Observe that

$$\beta = \frac{1}{T} \ !$$

We will see that this is generally true.

$$\boxed{\beta = \frac{1}{T} = \frac{\partial S_{eq}}{\partial U}.}$$